

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-V (NEW) EXAMINATION – WINTER 2021****Subject Code:3150910****Date:01/01/2022****Subject Name:Electrical Machine- II****Time:02:30 PM TO 05:00 PM****Total Marks: 70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		MARKS
Q.1	(a) Give technical reason of skewed rotor bar conductor of induction motor.	03
	(b) Explain working of 3-phase induction motor	04
	(c) Discuss the effect of harmonics, cogging, crawling and unbalanced voltages on performance of induction motor.	07
Q.2	(a) Why single phase induction motor not self started?	03
	(b) Explain working of capacitor start capacitor run single phase motor.	04
	(c) Explain working of Double cage induction motor	07
	OR	
	(c) What is regulation of alternator? Explain ZPF method for finding regulation in alternator.	07
Q.3	(a) Write advantages and application universal motor	03
	(b) Explain armature reaction phenomena in case of synchronous machine.	04
	(c) Draw the construction and explain working of cylindrical rotor synchronous machine	07
	OR	
Q.3	(a) Give comparison of squirrel cage and wound rotor motors.	03
	(b) State the methods of starting of synchronous motor and explain any one of them.	04
	(c) What is slip in induction motor? Explain torque/slip characteristics of induction motor.	07
Q.4	(a) Discuss the conditions to be satisfied before a 3- phase alternator is synchronized with infinite bus.	03
	(b) Justify that to get high starting torque slip ring motor is best suitable.	04
	(c) State the condition of parallel operation of alternator. Explain working of capacitor-start capacitor-run 1-phase induction motor	07
	OR	
Q.4	(a) What do you mean by synchronous speed of induction motor? Why can not 3-phase induction motor run at synchronous speed?	03
	(b) Draw and explain power stages of an induction motor	04
	(c) Draw and explain V-Curve of synchronous motor.	07
Q.5	(a) What is magnetic levitation?	03
	(b) What is Auto Synchronous Motor? Where it is used?	04
	(c) A 3- ϕ induction motor is wound for 6 poles and is supplied from 50-Hz system and has a slip of 1% at no-load and 3% at full load. Find (i) the synchronous speed (ii) no-load speed (iii) full load speed and (iv) frequency of rotor current at standstill. (v) Frequency of rotor current at full-load.	07
	OR	
Q.5	(a) State advantages and applications of linear induction motor	03
	(b) Explain how stepper motor works?	04
	(c) Discuss working and applications of Permanent magnet brushless DC motor	07